False hearing and L2-accented sentences

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Background

When bottom-up information is compromised, listeners may increase reliance on top-down information.

One way to assess such reliance is with the **false hearing paradigm**: Sommers et al. (2015) presented subjects with congruent and incongruent sentences (e.g., “She took the bridge to cross the river/liver”), finding greater false hearing (i.e., reporting “river” when the stimulus was “liver”) in older adults than younger adults.

Key Question: Do listeners show greater reliance on top-down processing when listening to L2 vs. L1-accented speech?

Methods

Participants: ~175 young adults, L1-English, normal hearing, no Italian/Italian-accented English exposure

Speakers: Two male young adults; 1 L1 English speaker; 1 Italian-accented English speaker

Transcription Task: 30 congruent, 30 incongruent sentences, split evenly between speakers. Participants listened to each one were then prompted to transcribe the whole sentence.

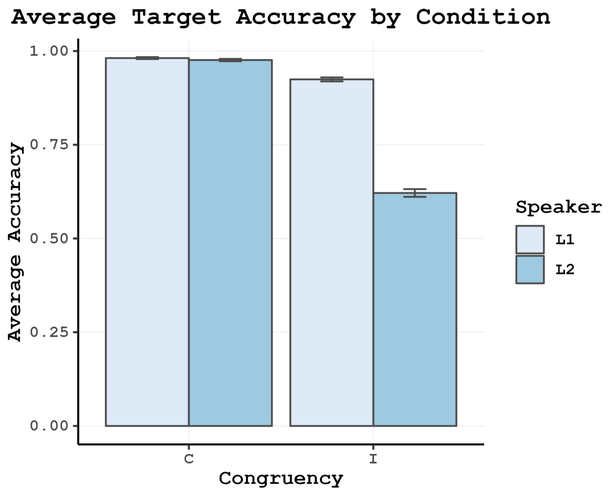
Confidence Ratings on a 1-10 scale provided by subjects after each trial, measuring their confidence in the accuracy of the final (target) word

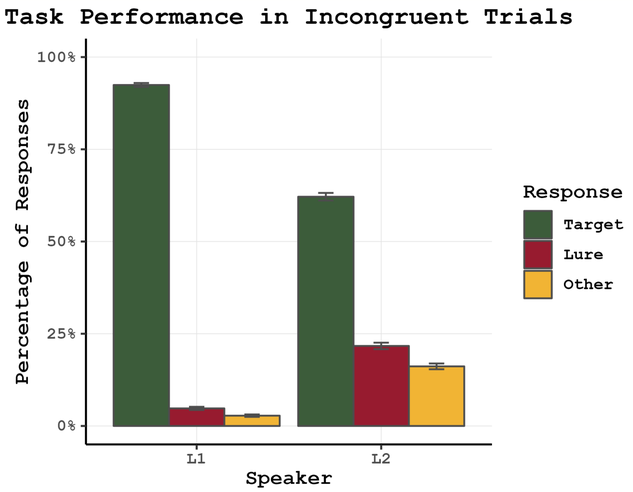
Tasks administered online using Gorilla

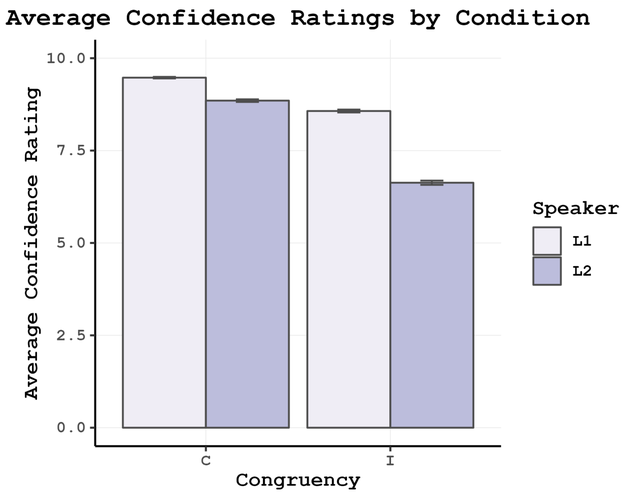
References

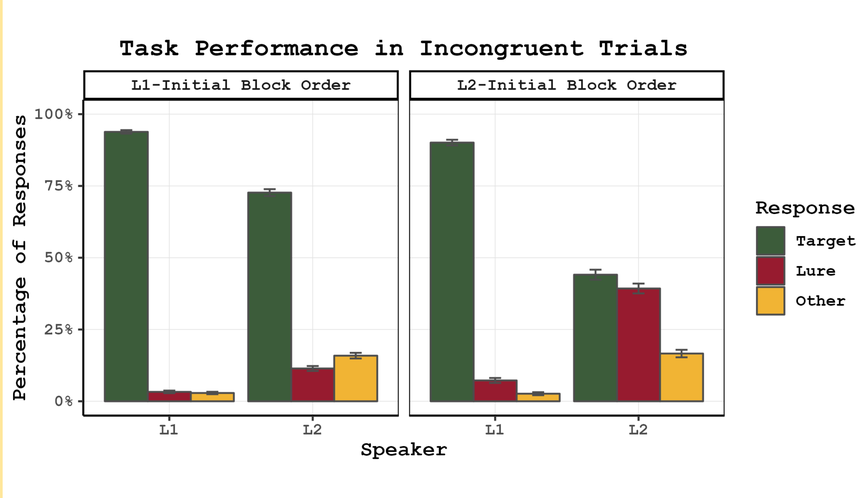
1. Van Engen & Peelle (2014)
2. Sommers et al. (2015)
3. Failes et al. (2020)
4. Rogers et al. (2012)
5. Schmid & Yeni-Komshian (1999)

Results









Discussion

- Young adults exhibited greater false hearing when listening to an L2-accented speaker than an L1-accented speaker, suggesting a greater reliance on top-down information when listening to L2-accented speech.
- Listeners reported higher subjective confidence ratings for their responses to the L1-speaker than the L2-speaker and higher confidence for congruent sentences over incongruent ones. Interestingly, their levels of confidence were similar for congruent L2 speech and incongruent L1 speech.
- False hearing for L2 speech was significantly higher when it was the first block listeners received. That is, when they don’t have reason to expect an L2 speaker to produce incongruent speech, they are more likely to rely on top-down processing.

- **Future Directions:** investigating effects of syllable position and/or confusability of phonological neighbors on false hearing

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